1. Introduction to Cybersecurity

- Basic Concepts: Confidentiality, Integrity, Availability (CIA Triad)
- Security Terminology: Threats, Vulnerabilities, Attacks
- Types of Cybersecurity: Network, Information, Application, and Cloud Security
- Cybersecurity Governance and Compliance
- Risk Management and Assessment

2. Networking Fundamentals

- Networking Basics: OSI Model, TCP/IP, Routers, Switches
- IP Addressing (IPv4 & IPv6), Subnetting
- DNS, DHCP, and NAT
- Network Topologies
- VPNs and Tunneling Protocols (IPSec, SSL, PPTP)
- Wireless Networking (Wi-Fi Security, WPA2, WPA3)

3. Threats and Attacks

- Types of Cyber Threats: Malware (viruses, worms, ransomware, spyware)
- Social Engineering (Phishing, Spear Phishing, Baiting)
- Denial-of-Service (DoS) and Distributed DoS (DDoS) Attacks
- Man-in-the-Middle (MitM) Attacks
- SQL Injection and Cross-Site Scripting (XSS)
- Insider Threats and Advanced Persistent Threats (APT)
- Zero-Day Exploits

4. Incident Response & Forensics

- Digital Forensics: Evidence Collection, Chain of Custody
- Malware Analysis and Reverse Engineering
- Analyzing Network Traffic, Memory, and Disk Images
- Forensic Tools: EnCase, FTK, Autopsy, Wireshark
- Legal and Ethical Issues in Cybersecurity
- Incident Response Lifecycle (Preparation, Detection, Containment, Eradication, Recovery

5. Penetration Testing and Ethical Hacking

- Ethical Hacking Concepts

- Reconnaissance and Footprinting
- Vulnerability Scanning and Assessment
- Exploitation and Post-Exploitation
- Tools for Penetration Testing: Metasploit, Nmap, Burp Suite, Wireshark
- Reporting Penetration Test Findings
- Legal and Ethical Implications of Penetration Testing

6. Application Security

- Web Application Security: OWASP Top 10 (SQL Injection, Cross-Site Scripting, etc.)
- Secure Coding Practices
- Application Firewalls and Web Application Firewalls (WAF)
- Secure APIs and Web Services (OAuth, JWT

7. Malware analysis

- Introduction to Malware
- Malware Analysis Basics
- Static Analysis
- Dynamic Analysis
- Malware Analysis Tools
- Reverse Engineering Techniques

Learning:-

Basic Networking & Operating Systems: Start with networking and operating system fundamentals. Knowing how systems communicate and are structured is essential for understanding cybersecurity.

Security Basics: Learn about encryption, firewalls, and intrusion detection systems. Understanding basic protection mechanisms is critical to cybersecurity.

Security Tools and Techniques: Familiarize yourself with common tools (e.g., Wireshark, Metasploit, Burp Suite) and techniques used in penetration testing, malware analysis, and incident response.

Hands-On Labs: Practice in virtual environments or through platforms like Hack The Box, TryHackMe, or CyberSecLabs, where you can learn through real-world scenarios.